SUEUN CHO

sueun.dev@gmail.com
github.com/sueun-dev | sueun-dev.github.io | linkedin.com/in/sueun-cho-625262252

PROFILE

Software engineer, open-source contributor, and Northwestern M.S. student with experience in backend systems, data pipelines, and blockchain development. Contributed merged correctness fixes to Prometheus, Open Policy Agent, pgx, and Trivy. Focused on reproducible debugging, numerical accuracy, and reliable software. Research interests: Bitcoin transaction selection and mempool linearization, block-template optimization, and Nakamoto consensus.

EDUCATION

Northwestern University
Evanston, IL | Fall 2026 - Present
Master of Science in Computer Science

University of Maryland
College Park, MD | May 2026
B.S. in Computer Science; Minor in General Business | GPA: 3.66/4.00
Coursework: Computer Systems, Computer Architecture, Network Security, Cryptography in Blockchain, Formal Languages

OPEN SOURCE CONTRIBUTIONS

Open Source Contributor
Selected merged contributions, 2026

- **Prometheus:** Corrected native-histogram counter-reset detection that could cause rate and increase calculations to undercount; added regression cases for both affected iterator paths. PR #19367
- **pgx (Go PostgreSQL driver):** Eliminated precision loss in scientific-notation parsing by replacing an intermediate float64 conversion with exact decimal parsing and exponent validation. PR #2628
- **Open Policy Agent:** Fixed silent integer-sum overflow with checked addition and an arbitrary-precision fallback, retaining the ordinary-integer fast path. PR #8987
- **Trivy:** Preserved dependency-graph edges for Composer packages incorrectly classified as PHP platform extensions; incorporated maintainer feedback and regression coverage. PR #11051

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, JavaScript, Java, Solidity, Rust
Tools: PostgreSQL, Docker, Git, GCP, AWS, Node.js, React, Web3.js
Focus: Backend development, data pipelines, regression testing, numerical correctness, Ethereum/EVM and Solana smart contracts

WORK EXPERIENCE

Chief Technology Officer, Blockchain Department | B&G Systems
Seoul, South Korea | May 2021 - Jul 2023

- Led a six-member engineering team building blockchain products; contributed to a 15% increase in company revenue.
- Designed smart contracts and backend systems, introduced design reviews and code audits, and delivered technical training to 10+ employees.

WORK EXPERIENCE - CONTINUED

Seasonal Freelance Blockchain Developer | Kmong

Jan 2020 - Present

- Developed and deployed 10+ smart contracts and token systems on Ethereum and Solana; designed on-chain architectures and API integrations for clients.

TECHNICAL PROJECTS

Co-founder | CoinHelio

Remote | Nov 2025 - Present

- Built real-time data pipelines aggregating cross-exchange signals for quantitative analysis and strategy testing.

Alcohol Prevention Research Chatbot | University of Maryland

Jan 2025 - Feb 2026

- Built backend systems for an NLP-based research chatbot on GCP, supporting reliable data collection and data integrity.

Co-founder | The-Adena, Blockchain Online Game

Seoul, South Korea | Mar 2022 - Dec 2023

- Led development of a blockchain game with 3,000+ users and $170K in revenue; implemented smart contracts and token systems with security validation.

Project Lead | Gomz Club, NFT Donation Project

Seoul, South Korea | Apr 2022 - Jul 2023

- Launched an NFT donation campaign featuring a public figure; raised and donated 1.25 ETH to Daon Rescue, an animal welfare organization, and secured national media coverage.

Kernel Optimization Challenge

Feb 2026

- Reached 1,146 cycles on Anthropic's original performance take-home, below every Claude reference result in its README (best: 1,363), through instruction scheduling, pipelining, and memory-access optimization.

LEADERSHIP & VOLUNTEER WORK

Developer | University Church

College Park, MD | Mar 2025 - Present

- Built and maintain a Flask/PostgreSQL platform supporting community operations.

Head of Research | Blockchain at Maryland

Feb 2025 - May 2026

- Led Web3 research initiatives and organized technical workshops.

Tech Leader | Korean Undergraduate Student Association, UMD

Aug 2024 - May 2026

- Built the organization's first official website and led technical initiatives.

TECHNICAL WRITING

Introduction to Smart Contracts with Solidity

Unpublished manuscript

- Sole author of a 260-page Korean manuscript on smart contract security and optimization.

AWARDS & HONORS

Dean's List: University of Maryland, Fall 2023 - Fall 2025

OMSE Academic Excellence Society: University of Maryland

Smart Contract Excellence Award: First place; developed a contract with 13% lower gas usage.

Blockchain Hackathon: Silver Prize, 2019